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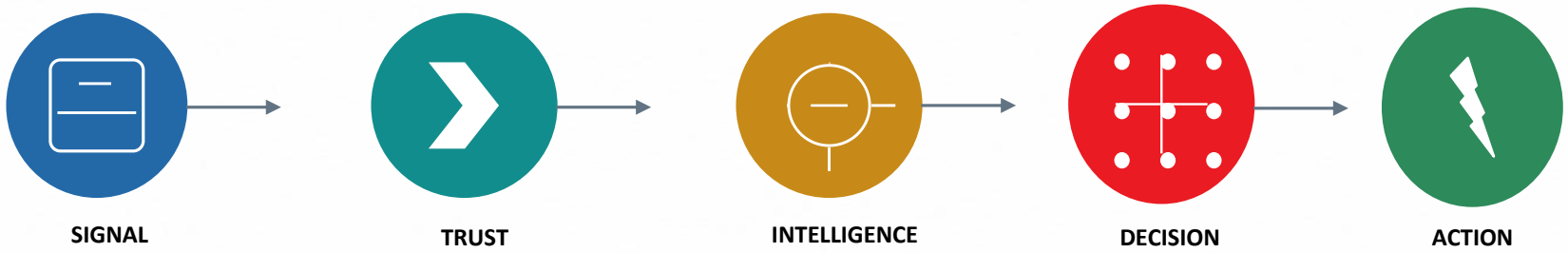
DUM 2026 CONFERENCE SESSION

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SCALING DIGITAL INTELLIGENCE: Advanced Smart Meter Data Analytics

Advanced Metering: From Signal to Action



Presented By

Santhosh S Nair, Director, Oracle Energy & Water

Scale is already here — and the data exhaust is accelerating

The analytics opportunity is growing faster than traditional utility operating models.

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Global Scenario

1B+
smart power meters worldwide (2022)

320M
distribution sensors deployed globally

13B
connected devices with controls/sensors (2023)

25B+
connected devices projected by 2030

India has the data. The next frontier is intelligence.

The rollout is large enough that the Narrative shifts from deployment to value realized from the signal.

72.4 Mil
smart meters installed across India as of 30 June 2026 (57.3 Mil in RDSS)

203.3 Ml
smart meters sanctioned under RDSS (consumer + DT + feeder scope)

₹1.31 lakh Cr
smart-metering works sanctioned under RDSS (Mar 2026)

12–15%
RDSS target range for pan-India AT&C losses

19.2 B Data points
96 interval data /day/channel from 200 Mil+ smart mdeters

Why this matters

Higher meter penetration means more opportunities for pattern detection, customer segmentation, outage intelligence and distribution visibility — but also more data quality and governance pressure.

Are we realizing value from communicating meters ?

COMMUNICATION	How many meters are actually communicating?
TRUST	How much interval data is usable and trusted?
ACTION	How many insights trigger a workflow?
VALUE	What business outcome is measured ?

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Source: https://www.tndindia.com/indias-smart-meter-population-at-7-24-crore-parliament/
Sources: IEA (2023); U.S. EIA Electric Power Annual, Table 10.05 (data for 2024); Ministry of Power / PIB (Jan–Aug 2026).

A Day at Ujjwal DISCOM Smart Meter Operation Center -- Illustrative

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Global smart-meter journey: deployment → analytics → active grid

Across leading markets, the strategic journey is moving from meter installation to operational, customer and flexibility use cases.

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1 — AMI SCALE

Advanced metering infrastructure becomes broadly deployed

→

2 — DATA TRUST

VEE, device health, topology and time-aligned interval data mature

→

3 — OPERATIONS

Outage, loss, asset and planning analytics become everyday workflows

→

4 — CUSTOMER

Usage insights, TOU, targeted messaging and programme engagement grow, Bill optimization lead to savings

→

5 — FLEXIBILITY

DER, EV, storage and demand response, flexibility become measurable and orchestrated

From rollout economics to intelligence economics

Value realization: cash & service now. → predictive operations → flexibility and portfolio steering later.

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0–12 MONTHS

PROTECT • RESTORE • ENGAGE

- Revenue assurance
- Billing / collection exceptions
- Outage intelligence
- Proactive customer journeys

Observable KPI

AT&C • collection • outage minutes • call volume

1–3 YEARS

PREDICT • TARGET • OPTIMISE

- Predictive meter & asset health
- Loss-risk targeting
- Peak / DER forecasting
- Targeted demand response

Observable KPI

forecast error • exception closure • participation

3–5+ YEARS

FLEX • ORCHESTRATE • STEER

- Flexible-load valuation
- DER / EV / storage orchestration
- Power procurement optimisation
- Generation portfolio steering

Observable KPI

peak MW • procurement cost • avoided capacity

THE ECONOMIC ARC

more trusted data → better prediction → better decisions → measurable outcomes

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AI/ML is the multiplier — but readiness is the gate

REC's 2026 scale-up framework reinforces: choose use cases by digital readiness, baseline outcomes, validate, then scale.

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METER / LOSS

Anomaly scoring • VEE prioritisation • loss patterns

Action: prioritise investigations by value-at-risk

OUTAGE / RELIABILITY

Last-gasp clustering • restoration verification • crew prioritisation

Action: improve restoration decisions in OMS / field workflows

DEMAND / FLEXIBILITY

Peak forecasting • EV/solar/storage signatures • flexibility propensity

Action: target DR / DER programs and quantify delivered flexibility

READINESS GATES BEFORE AI/ML

DATA TRUST	CONTEXT	WORKFLOW	GOVERNANCE	MEASUREMENT
completeness • VEE • timeliness	topology • IDs • time alignment	owner • threshold • execution path	privacy • security • drift • override	baseline • target • realised impact

REC framing: Revenue Creation + Revenue Optimisation buckets • independent validation • KPI baselining • scale-up after pilot validation

Long run: digital intelligence can turn flexible load into a grid resource

The strategic opportunity is to move from forecasting demand to valuing, orchestrating and verifying flexibility.

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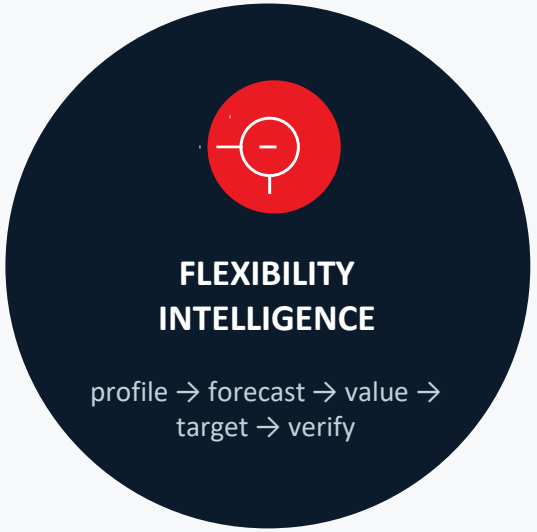
FLEXIBILITY SOURCES

C&I load
shift / shed

EV charging
managed charging

Solar + storage
shift / firm

Residential load
behavior / TOU



FROM FLEXIBILITY TO PORTFOLIO VALUE

Demand response	target programmes / aggregators
Managed charging	shape EV load around system needs
Storage shifting	move energy across time windows
Procurement optimisation	reduce peak exposure and deviation
Generation portfolio	inform renewable / firming / conventional mix decisions

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Build intelligence now — design for the India Energy Stack

IES is emerging as the interoperability rail; it should shape architecture choices without becoming a prerequisite for value delivery.

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WHERE IES IS

Emerging, not yet nationwide

MoP has identified pilot utilities and a FY2026–27 demonstration window.

Implication

Do not wait for a national stack. Build AI/ML capabilities now with open interfaces, common identifiers, lineage and reusable data contracts.

Design principle

IES-ready ≠ IES-dependent



INFORMATION Common identifiers • data exchange • standard protocols



SERVICES Reusable APIs • analytics / decision services • workflow access



TRUST Consent • security • lineage • accountable use of data



MONEY Settlement • flexibility • market / commercial transactions

Oracle energy solutions play in Signal to Action ...

Anchors for a utility AI/ML business case.

1 Customer Engagement

DATA SIGNALS

Billig history, Customer profile, Meter interval data , Account info, , Program info, Tariff plan, DER info, Appliance level Disagg. ..

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Opower

ACTION Proactive alerts, Personalized comms, new program intro, Speciales rate plans, Demand response communications, Self service enhancemet

MEASURE

, Peak load Energy Saving, reduce Cost to Serve, faster Program adoption, Energy Awareness

2 Loss intelligence

DATA SIGNALS

AMI interval data + meter events + billing + network balance

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Analytics Insights / MDM

ACTION Risk-ranked inspection queue

MEASURE

AT&C / non-technical losses

3 Outage intelligence

DATA SIGNALS

SCADA + AMI last gasp + topology + weather + history

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NMS / ADMS + OMS

ACTION Fault inference + restoration plan

MEASURE

SAIDI / SAIFI / MTTR

4 Asset risk

DATA SIGNALS

Condition + loading + age + outages + weather

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Data Intelligence + WAM/APM

ACTION Maintenance / replace priority

MEASURE

Failure rate / capex

4 DER / net-load

DATA SIGNALS

AMI + weather + DER telemetry + EV adoption

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DERMS + NMS / ADMS

ACTION Constraint-aware dispatch / reinforcement trigger

MEASURE

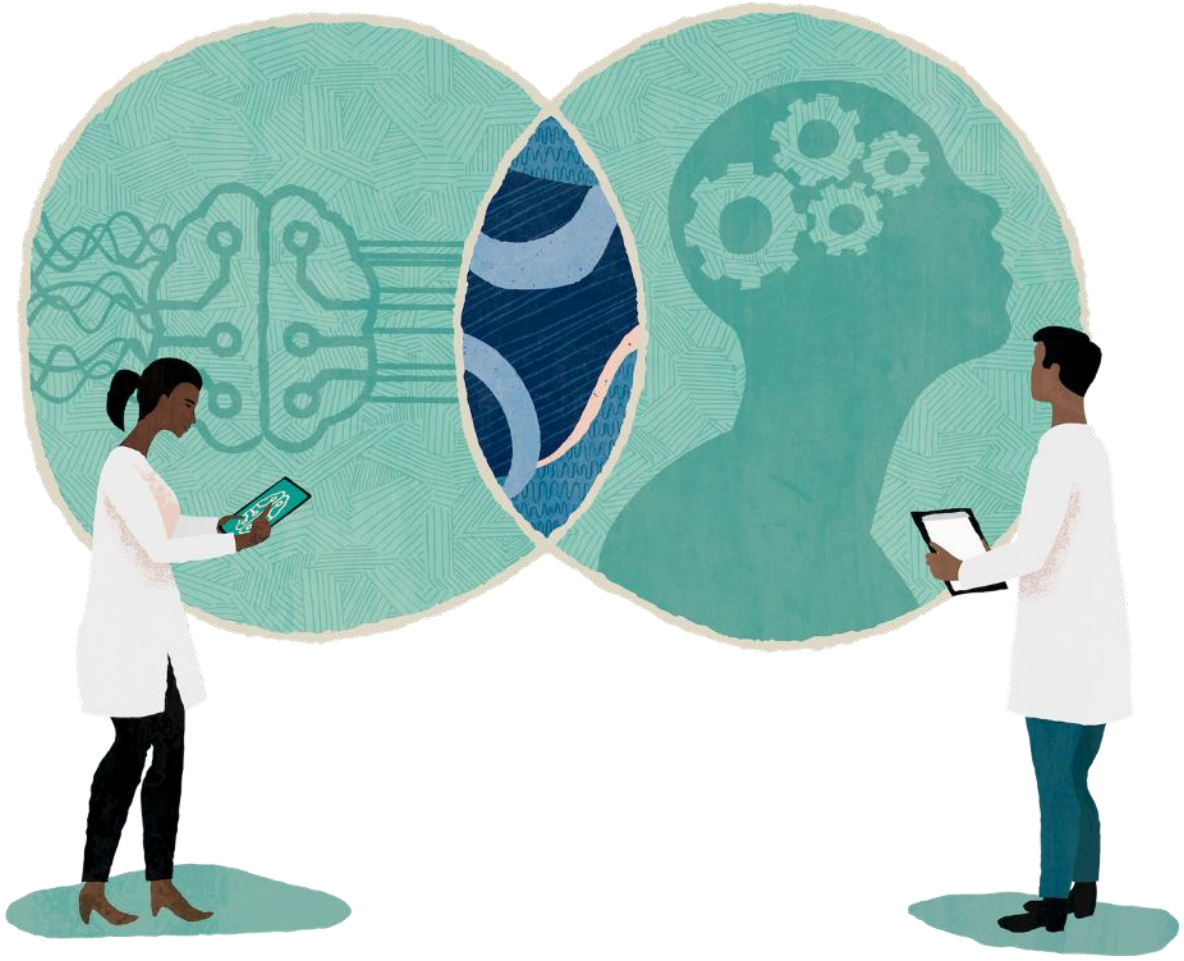
Hosting capacity / capex

GenAI sits above these plays as the interface: explain the prediction, summarize evidence, generate work instructions, and capture field feedback.

Influencing customer action across 175+ Utilities across Globe through AI and Behavioural Science – Oracle Utilities/Opower does this…

Artificial Intelligence

- Simplifies complexity
- Trained on the world's largest energy consumer data set
- Predicts and offers what each customer needs



Behavioural Science

- A variety of proven principles
- Large experiments with leading scientists
- Gets customer attention and influences action

Oracle Utilities- Business Impacts towards different Utility Goals

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Energy Awareness, Optimization & Engagement

BSES Rajdhani Power Limited

Objective: Leverage behavioral approach to increase Energy Awareness among consumers, customer satisfaction and brand recognition

17% reported reduced calling

82% liked HERs

Customer Engagement

Mercury

Objective: Increase band value and reduce customer churn in highly competitive market

5% Brand Value

76% Customer Engagement

19% Reduction High Bill Calls

Scaled Adoption

PG&E Pacific Gas and Electric Company®

Objective: Develop a long-term strategy for the end-to-end customer self-service experience across digital channels for all customer segments

40% avg time on site

450k Next Best Actions taken

Automation

conEdison

Objective: Refresh the online customer experience with a focus on simplicity, personalization, and ease-of-use

29% more purchased overall on Marketplace

61% product sales

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Oracle has public utility references across various operational domains

SMECO Electric / US	**Oracle Utilities Analytics Insights**	Used Smart meter data to identify 50 overloaded transformers in first few months that could have caused unplanned outages	**Outage / Asset Health**
Xcel Energy Electric / US	**Oracle Utilities NMS**	Minnesota control center avoided ~934 invalid outage calls, translating to ~1,400 work hours, using meter “ping” verification.	**Outage / field/Meter health**
American Electric Power (AEP) Electric / US	**Oracle Utilities Opower + Oracle Utilities platform**	AEP Ohio had already delivered >25M personalized communications, helping customers save millions; broader platform modernization spans customer and meter-to-cash.	**Customer / AMI**
Eversource Electric / US	**Oracle Utilities OMA + NMS**	Mobile damage assessment and potential restoration workflows integrated with NMS during major events.	**Storm / field**
Louisville Water Water / US • adjacent	**Oracle Utilities WAM / APM**	Forecasting asset cost and doing predictive maintenance before failures to extend asset life.	**Asset / APM**
NIE Networks Electric / UK	**Oracle Utilities NMS**	ADMS for outage + grid operations; improved reliability/restoration; foundation for power flow and DER management.	**Outage / DER**

These references validate product/process adoption; they should not be read as proof that each exact use case used ML.

Key takeaways

The strategic asset is not the meter. It is the reusable intelligence loop around the meter.

- **Scale the decision, not the dashboard**
  - Analytics has value when an anomaly becomes a prioritized work item, customer action, switching decision or program intervention.
- **Utility context is the multiplier**
  - Meter data becomes materially more useful when joined with topology, customer, asset, outage, weather and field context.
- **Productize the repeatable 80%; extend the differentiating 20%**
  - Use utility-native capabilities for governed meter and operational semantics, then AI for utility-specific models and new use cases.
- **Close the loop**
  - A model score stranded in BI is not an operating outcome. Embed intelligence into OMS, CIS, work/asset, field and customer workflows.
- **Measure intelligence economics**
  - Attach each use case to AT&C loss, collection efficiency, SAIDI/SAIFI, avoided truck rolls, call volume, field productivity or program participation.

THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

Links/References (If any)

sant.nair@oracle.com



Discussion prompts

1. Where is the highest-value meter signal today — and why is it not yet operationalized?
2. Which 3 use cases could prove measurable value in the next 90 days?
3. What data-quality or integration bottleneck would block scaling from 1–3 use cases to 10–20?
4. Where should AI assist, where should humans decide, and where should automation act?
5. How do utilities avoid building a new analytics silo for every product or program?

A pragmatic path: prove value, industrialise the pattern, then scale

Leave reusable assets behind at every stage.

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0–90 DAYS

PROVE



3 use cases • baseline KPI • data heatmap • pilot

Gate: measurable value + reusable capability

3–6 MONTHS

INDUSTRIALISE



reusable features • semantic layer • APIs • governance

Gate: measurable value + reusable capability

6–12 MONTHS

SCALE



10–20 use cases • cross-domain decision products • value ledger

Gate: measurable value + reusable capability

12–18 MONTHS

CONNECT



closed-loop automation • IES-ready interfaces • flexibility optimisation

Gate: outcomes + IES-ready operating pattern

The strategic asset is not the meter. It is the reusable intelligence loop around the meter.

Oracle Utilities- Opower Customer Engagement Journey

